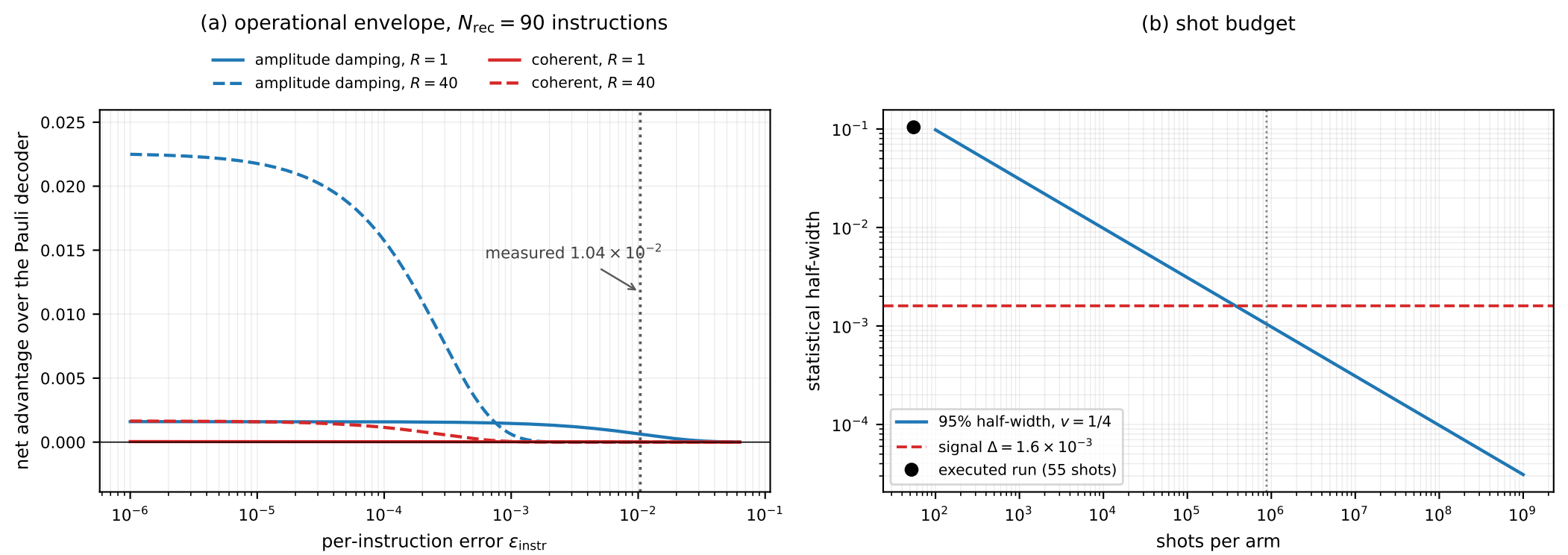
(a) operational envelope, $N_{\mathrm{rec}} = 90$ instructions
amplitude damping, $R = 1$
amplitude damping, $R = 40$
coherent, $R = 1$
coherent, $R = 40$
net advantage over the Pauli decoder
0.025
0.020
0.015
0.010
0.005
0.000
measured 1.04×10^{-2}
10^{-6}
10^{-5}
10^{-4}
10^{-3}
10^{-2}
10^{-1}
per-instruction error $\varepsilon_{\mathrm{instr}}$
(b) shot budget
statistical half-width
10^{-1}
10^{-2}
10^{-3}
10^{-4}
95% half-width, $v = 1/4$
signal $\Delta = 1.6 \times 10^{-3}$
executed run (55 shots)
10^{2}
10^{3}
10^{4}
10^{5}
10^{6}
10^{7}
10^{8}
10^{9}
shots per arm